

DATA SCIENCE INTERNSHIP

MINOR PROJECT REPORT

Exploratory Data Analysis & Insights on Student Performance

Prepared By

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Data Science Internship

August 2026

Project Focus: Data Cleaning • Exploratory Data Analysis • Visualization • Predictive Analysis • Insight Generation

This report documents the analysis implemented in the project repository, including the notebooks, raw datasets, preprocessing workflow, model evaluation, and visual evidence.

Acknowledgement

I would like to express my sincere gratitude to everyone who supported me during the completion of this Data Science internship project. This project provided an opportunity to apply Python-based data analysis techniques to an educational dataset and strengthened my understanding of data cleaning, exploratory analysis, visualization, preprocessing, and predictive modeling.

I am also thankful to the internship program for providing the project framework and learning environment that encouraged practical implementation. Finally, I appreciate the availability of the UCI Student Performance dataset, which made it possible to investigate factors associated with academic outcomes.

Abstract

This project investigates student academic performance using the UCI Student Performance dataset. The analysis focuses primarily on the Mathematics dataset, containing 395 student records and 33 attributes, with final grade (G3) treated as the main outcome variable. The workflow includes dataset inspection, missing-value and duplicate checks, descriptive statistics, univariate analysis, bivariate analysis, visualization, categorical encoding, feature scaling, and supplementary regression modeling.

The exploratory analysis showed a broad distribution of final grades, while the modeling notebook extended the analysis by combining the Mathematics and Portuguese datasets, removing duplicate demographic records, and evaluating Linear Regression, Decision Tree, and Random Forest regressors. Random Forest produced the strongest recorded R^2 score of 0.813. Feature-importance analysis indicated that G2 was the most influential predictor in the fitted Random Forest, followed by absences and other student attributes.

The project demonstrates how raw educational records can be transformed into interpretable evidence for academic monitoring and intervention. The findings should be interpreted as associations and predictive relationships rather than causal effects.

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The report follows the requested internship-report structure and the actual workflow found in the project repository.

1. Introduction

Educational institutions generate large amounts of student information covering academic performance, attendance, family background, study habits, social activities, and other characteristics. Raw records, however, are not automatically useful for decision-making. Data analysis can reveal patterns that help educators understand student performance and identify areas where additional support may be appropriate.

This project applies a practical data-science workflow to the UCI Student Performance dataset. The repository contains separate notebooks for exploratory analysis and preprocessing/modeling, together with Mathematics and Portuguese raw CSV files. The EDA notebook analyzes the Mathematics dataset, while the second notebook extends the work to multi-subject preprocessing and predictive modeling.

2. Problem Statement

Educational institutions may possess detailed student records but still struggle to convert those records into actionable insights. The problem addressed in this project is to explore student-level academic data systematically, identify patterns associated with final grades, visualize relationships between relevant variables, and demonstrate how predictive models can support interpretation.

3. Objectives

- Understand the structure, variables, and target outcome of the student-performance data.
- Check data quality through missing-value and duplicate analysis.
- Perform univariate, bivariate, and multivariate exploratory analysis.
- Visualize student-grade distributions and relationships involving study time and absences.
- Transform categorical variables into machine-readable features using one-hot encoding.
- Apply feature scaling where required for the modeling workflow.
- Compare regression models using MAE, RMSE, and R^2 .
- Identify influential predictors using Random Forest feature importance.
- Translate analytical results into plain-language educational recommendations.

4. Dataset

4.1 Source and Structure

The project uses the UCI Student Performance dataset. The repository stores two raw files: *student-mat.csv* for Mathematics and *student-por.csv* for Portuguese. The Mathematics dataset used in the EDA notebook contains 395 rows and 33 columns. Variables cover demographic information, family and educational background, study habits, social activities, health-related self-reported attributes, absences, and grade variables G1, G2 and G3.

Aspect	Project detail
Primary EDA dataset	student-mat.csv (Mathematics)
EDA size	395 rows × 33 columns
Target variable	G3 – final grade (0–20)
Grade variables	G1, G2, G3
Raw companion dataset	student-por.csv (Portuguese)
Combined modeling data	662 rows after cross-course duplicate removal
Encoded combined features	42 features

4.2 Important Variables

Academic: G1, G2, G3. **Attendance:** absences. **Study behavior:** studytime, failures, paid classes, school support and related variables. **Demographic/family:** sex, age, address, family size, parental education and jobs. **Lifestyle:** free time, going out, alcohol consumption, health and related attributes.

The EDA notebook records no missing values and no duplicate rows in the Mathematics dataset. Final-grade statistics were mean = 10.42, median = 11, standard deviation = 4.58, minimum = 0, and maximum = 20.

5. Tools & Technologies

Technology	Use in project
Python	Core programming and analysis language
Pandas	Data loading, inspection, transformation and tabular analysis
NumPy	Numerical operations and metric calculations
Matplotlib	Plotting and chart presentation
Seaborn	Statistical visualization and distribution plots
Scikit-learn	Encoding/scaling workflow, train-test split, regression models and metrics
Jupyter Notebook / VS Code	Interactive development and experiment tracking
GitHub	Version control and project hosting

6. Methodology

The project follows a standard data-science pipeline: data acquisition → data understanding → data-quality checks → exploratory analysis → visualization → preprocessing → model development → evaluation → feature interpretation → recommendations.

Stage	Implementation
Data loading	CSV files loaded with Pandas using semicolon separators.
Data understanding	Shape, first rows, variables and G3 descriptive statistics examined.
Quality checks	Missing values and duplicate rows checked.
EDA	Grade distribution, gender distribution, study time and absences examined.
Preprocessing	Categorical variables one-hot encoded; G3 separated as target.
Scaling	StandardScaler applied to train/test feature matrices.
Modeling	Linear Regression, Decision Tree and Random Forest regressors evaluated.
Interpretation	MAE, RMSE, R^2 and Random Forest feature importance used.

7. Data Cleaning & Preprocessing

7.1 Initial Quality Checks

The EDA notebook loads the Mathematics dataset and checks dimensions, missing values, duplicate rows and final-grade statistics. The recorded result is 395 rows and 33 columns, with zero missing values and zero duplicate rows. No imputation step was required for the EDA dataset.

7.2 Combined-Dataset Preparation

For the supplementary modeling workflow, both raw datasets were loaded and a subject indicator was added before concatenation. Duplicate student records were removed using demographic columns. The resulting combined

dataset contained 662 rows. Categorical variables were transformed using one-hot encoding and G3 was separated as the target.

7.3 Train-Test Split and Scaling

The combined workflow used an 80/20 train-test split, producing 529 training rows and 133 testing rows after preprocessing. The Mathematics-only modeling section used 316 training rows and 79 testing rows. StandardScaler was fitted on the training data and applied to both training and test features.

Important interpretation: G1 and G2 are earlier academic grades and are naturally highly informative about G3. Their predictive importance should not be interpreted as proof that they independently cause the final grade.

8. Exploratory Data Analysis

8.1 Univariate Analysis – Final Grade

The distribution of G3 is centered around the middle of the 0–20 grading scale, with a substantial concentration around approximately 10–15. The recorded descriptive statistics show a mean of 10.42 and median of 11. The distribution also contains students at the lower end of the scale, including zero final-grade observations.

8.2 Gender Distribution

The gender-count visualization shows a moderately higher number of female students than male students in the Mathematics dataset. This is descriptive and does not by itself establish any relationship between gender and academic performance.

8.3 Bivariate Analysis

The EDA notebook examines weekly study time against final grade using a box plot and school absences against final grade using a scatter plot. These plots help identify differences across study-time categories and visually inspect the relationship between absences and G3.

9. Visualizations

The following figures are based on the executed project notebooks and the screenshots supplied with the project.

9.1 Final Grade and Gender Distribution

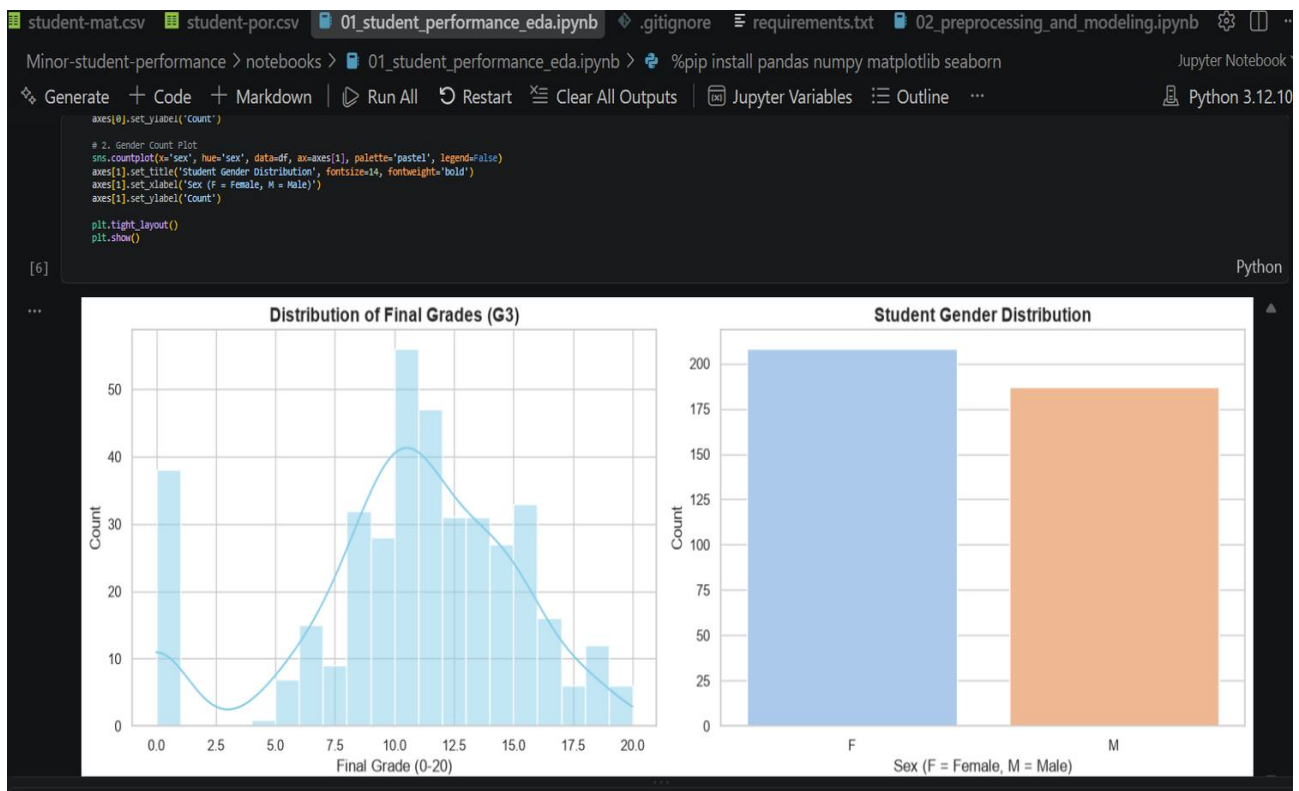


Figure 1. Distribution of final grades (G3) and student gender counts from the executed EDA notebook.

9.2 Random Forest Feature Importance

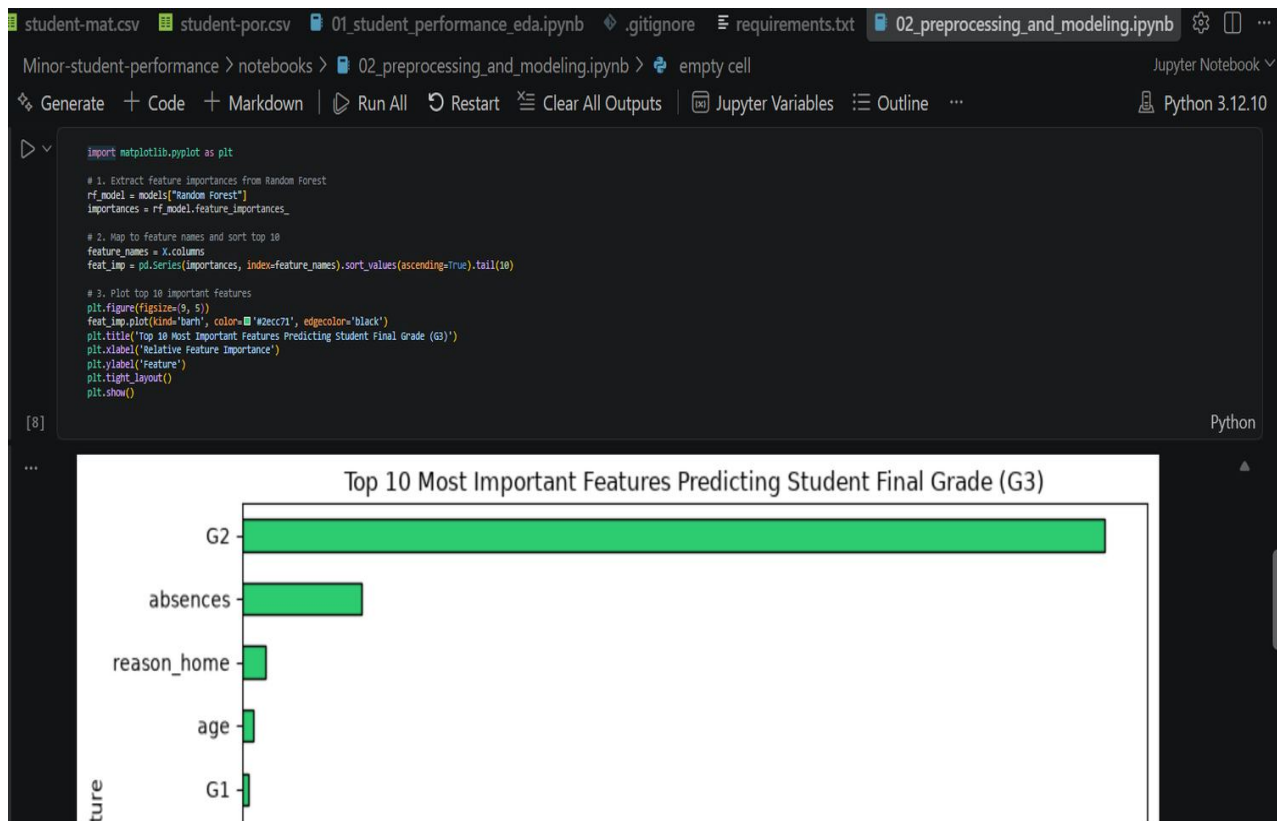
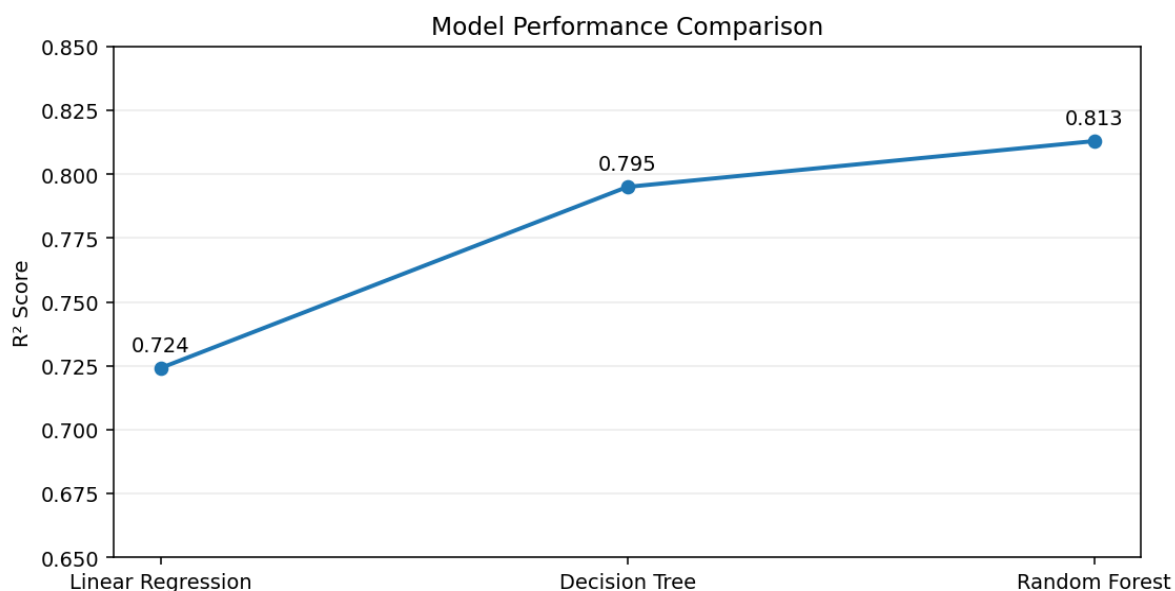


Figure 2. Top features identified by the Random Forest model. G2 is visibly dominant, followed by absences and other features.

10. Predictive Analysis & Model Evaluation

Although the project brief emphasizes exploratory data analysis, the repository also contains a second notebook that performs supplementary predictive analysis. Three regression algorithms were trained and evaluated: Linear Regression, Decision Tree and Random Forest.

Model	MAE	RMSE	R ² Score
Linear Regression	1.647	2.378	0.724
Decision Tree	1.139	2.050	0.795
Random Forest	1.179	1.958	0.813



Random Forest achieved the highest R² score (0.813) and the lowest RMSE (1.958) among the three recorded models. Decision Tree achieved the lowest MAE (1.139).

10.1 Feature Importance

The Random Forest model was used to calculate relative feature importance. The recorded top-10 visualization places G2 far above the remaining features, followed by absences and several demographic/academic variables. This indicates that the fitted model relies strongly on recent academic performance when predicting G3.

11. Results

- The Mathematics EDA dataset contains 395 students and 33 attributes.
- No missing values or duplicate rows were found in the Mathematics EDA dataset.
- The average Mathematics final grade (G3) is approximately 10.42/20, with a median of 11.
- The final-grade distribution is concentrated around the middle of the grading scale but includes low and high performers.
- Female students outnumber male students in the analyzed Mathematics records.
- Study time and absences were selected for direct bivariate visual analysis against G3.
- The combined multi-subject preprocessing workflow produced 662 rows after duplicate removal and 42 encoded features.
- Random Forest produced the strongest recorded R² score of approximately 0.813.
- G2 is the dominant feature in the Random Forest importance analysis.

12. Key Insights

Insight 1 – Previous academic performance matters strongly

G2 dominates the Random Forest feature-importance chart. This is expected because G2 is an earlier grade from the same academic process and is therefore a strong indicator of a student's current performance trajectory.

Insight 2 – Attendance deserves attention

Absences appear among the most important non-grade features in the Random Forest output. This suggests attendance-related information can be useful for identifying students who may require closer academic monitoring.

Insight 3 – Study behavior is worth monitoring

The project directly visualizes study time against G3. While the analysis does not establish causality, study behavior can be incorporated into a broader student-support framework.

Insight 4 – A single variable should not define student success

The modeling results show that several variables contribute to predictions even though G2 is much more influential than the others. Educational decisions should therefore consider multiple indicators.

13. Business / Educational Recommendations

- Develop an early-warning dashboard using recent grades and attendance indicators to flag students who may need academic support.
- Use periodic assessment results to monitor performance trajectories instead of waiting until the final examination.
- Track absence patterns and consider timely outreach when attendance begins to deteriorate.
- Combine academic indicators with contextual variables so interventions are supportive rather than purely score-driven.
- Use model predictions as decision-support evidence, not as automatic judgments about students.
- Periodically retrain and validate models when new institutional data become available.

14. Limitations

- The dataset represents students from specific Portuguese schools and may not generalize to every educational setting.
- The EDA primarily analyzes Mathematics, while the supplementary modeling workflow combines Mathematics and Portuguese records.
- G1 and G2 are temporally close to G3, making them highly informative but limiting their usefulness for truly early prediction.
- Random Forest feature importance describes model reliance, not causal influence.
- The project uses a single train-test split rather than cross-validation, so metrics may vary with a different split.
- Some variables are self-reported or categorical proxies for complex social and educational conditions.

15. Future Scope

- Use cross-validation and hyperparameter tuning for more robust model comparison.
- Compare additional algorithms such as Gradient Boosting and regularized regression.
- Build early-prediction models that exclude G1/G2 to identify factors available before major assessments.
- Add explainability methods such as SHAP or permutation importance.
- Develop an interactive dashboard using Power BI, Streamlit or another visualization platform.
- Evaluate fairness and stability across demographic groups before operational use.

16. Conclusion

This minor project demonstrates an end-to-end approach to transforming educational data into meaningful analytical evidence. The workflow began with dataset inspection and quality checks, progressed through exploratory visualization, and was extended with preprocessing and regression modeling. The analysis found that the Mathematics dataset contained no missing or duplicate rows and that final grades were distributed broadly around the middle of the 0–20 scale.

The supplementary predictive analysis showed that Random Forest achieved the strongest overall recorded performance, with an R^2 of approximately 0.813 and RMSE of approximately 1.958. Feature importance highlighted G2 as the dominant predictor, while absences and other student characteristics provided additional information.

Overall, the project demonstrates the core data-science skill targeted by the internship: converting raw student records into clear visualizations, quantitative results, and practical recommendations. The findings are most appropriately used to support human decision-making and early academic intervention.

17. Work Evidence

The project repository contains the notebooks used for analysis. The screenshots included in this report provide visual evidence of the implemented work, including the EDA plots and Random Forest feature-importance analysis.

Evidence	Description
Notebook 01	Dataset inspection, quality checks, G3 statistics, distributions and bivariate plots.
Notebook 02	Multi-subject preprocessing, encoding, scaling, train-test split, model evaluation and feature importance.
Screenshot 1	G3 distribution and gender distribution.
Screenshot 2	Random Forest top-10 feature importance.
Raw data	student-mat.csv and student-por.csv stored in data/raw.

18. GitHub & References

Project Repository

<https://github.com/khushipspective-dot/Data-Science-Internship---Minor-Major-Projects>

Dataset Reference

<https://archive.ics.uci.edu/dataset/320/student%2Bperformance>

Project Files Reviewed

Minor-student-performance/notebooks/01_student_performance_eda.ipynb
Minor-student-performance/notebooks/02_preprocessing_and_modeling.ipynb
Minor-student-performance/data/raw/student-mat.csv
Minor-student-performance/data/raw/student-por.csv

Report preparation note: The report content and numerical results are based on the project repository, executed notebook outputs, and supplied screenshots available at the time of report generation.